

Campus Hiring Evaluation Submission

Name

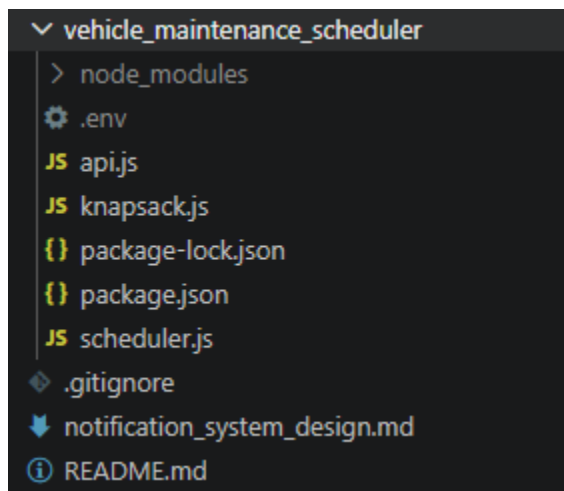
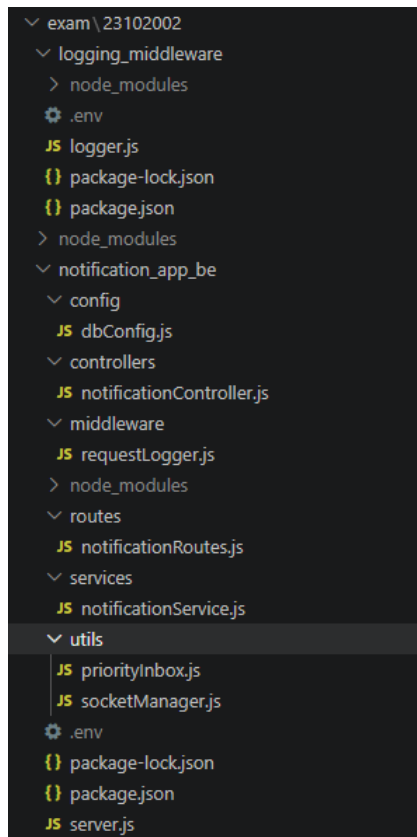
Dhawal Shukla

Roll Number

23102002

1. Project Folder Structure

Added complete backend project structure.



2. Logging Middleware

Implemented reusable custom logging middleware using the provided logging API.

Features:

- external log API integration
- reusable logger
- protected API logging

Files:

- logging_middleware/logger.js

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  POSTMAN CONSOLE

C:\Users\power.jiit\Desktop\Dhawal\exam\23102002\notification_app_be>npm start

> notification_app_be@1.0.0 start
> node server.js

◇ injected env (3) from .env // tip: % override existing { override: true }
Server running on port 3000
Invalid URL
Terminate batch job (Y/N)? y

C:\Users\power.jiit\Desktop\Dhawal\exam\23102002\notification_app_be>npm start

> notification_app_be@1.0.0 start
> node server.js

◇ injected env (3) from .env // tip: % auth for agents [www.vestauth.com]
◇ injected env (0) from .env // tip: % auth for agents [www.vestauth.com]
Server running on port 3000
{
  logID: 'b8b743f6-6519-45c5-a6fe-b063170b6093',
  message: 'log created successfully'
}
[]
```

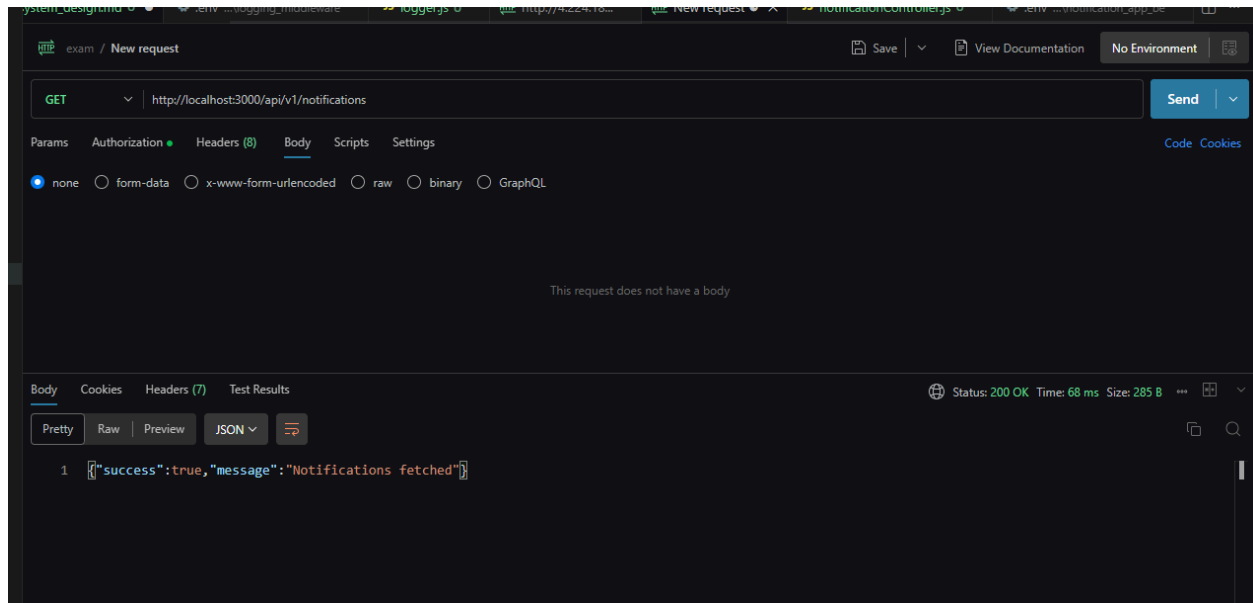
3. Notification Backend APIs

Implemented notification backend using Express.js.

APIs:

- GET /notifications
- GET /notifications/:id
- POST /notifications
- PATCH /notifications/:id/read
- PATCH /notifications/mark-all-read
- DELETE /notifications/:id

Protected routes added using bearer token middleware.



4. Vehicle Maintenance Scheduler

Implemented vehicle scheduling system using Dynamic Programming (0/1 Knapsack).

Features:

- fetched depots from API
- fetched vehicles from API
- calculated maximum impact within mechanic hours

```
◇ injected env (1) from .env // tip: % custom filepath { path: '/custom/path/.env' }  
Maximum Impact: 214
```

```
C:\Users\power.jiit\Desktop\Dhawal\exam\23102002\vehicle_maintenance_scheduler>node scheduler.js
```

```
◇ injected env (1) from .env // tip: % custom filepath { path: '/custom/path/.env' }
```

```
Depots: [
```

```
  { ID: 1, MechanicHours: 60 },  
  { ID: 2, MechanicHours: 135 },  
  { ID: 3, MechanicHours: 188 },  
  { ID: 4, MechanicHours: 97 },  
  { ID: 5, MechanicHours: 164 }
```

```
]
```

```
Vehicles Sample: [
```

```
  {  
    TaskID: 'f270081d-ae79-4ef8-bae1-af6c19689538',  
    Duration: 4,  
    Impact: 10
```

```
  },
```

```
  {  
    TaskID: '6498f532-2169-407c-9f4f-0035cd71b99c',  
    Duration: 8,  
    Impact: 4
```

```
  },
```

```
  {  
    TaskID: 'a62d7829-88f7-4741-abc4-bc5ed2c5849e',  
    Duration: 8,  
    Impact: 7
```

```
  }
```

```
]
```

```
Depot Mechanic Hours: 60
```

```
Maximum Impact: 153
```

```
C:\Users\power.jiit\Desktop\Dhawal\exam\23102002\vehicle_maintenance_scheduler>
```

5. Priority Inbox System

Implemented top 10 notification priority system.

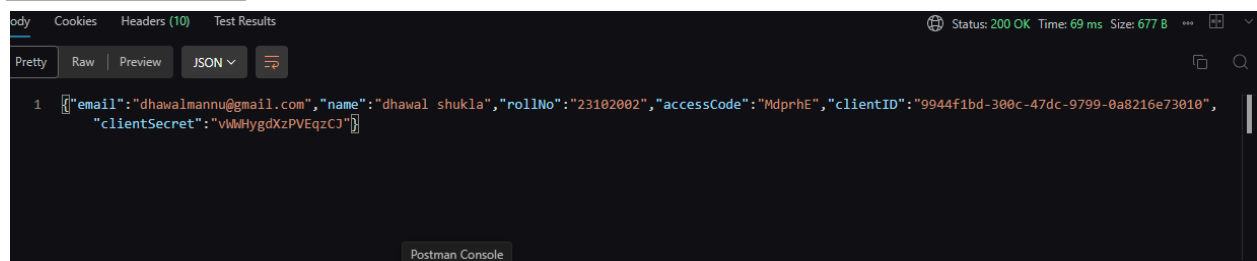
Priority Order:

Placement > Result > Event

Used sorting / priority based approach for ranking notifications.

```
C:\Users\power.jiit\Desktop\Dhawal\exam\23102002\notification app be>node utils/priorityInbox.js
injected env (3) from .env // tip: custom filepath { path: '/custom/path/.env' }
Top 10 Notifications:
[
  {
    ID: 'e4ef6295-1d3a-4b8c-86d9-4942ea7f4ec2',
    Type: 'Placement',
    Message: 'Booking Holdings Inc. hiring',
    Timestamp: '2026-05-08 08:46:13'
  },
  {
    ID: '1576dd1c-4729-4645-b603-155852ad93f4',
    Type: 'Placement',
    Message: 'Broadcom Inc. hiring',
    Timestamp: '2026-05-08 01:45:45'
  },
  {
    ID: '92c6174f-8107-4f9b-8abd-aa7259ecffeb',
    Type: 'Placement',
    Message: 'TSMC hiring',
    Timestamp: '2026-05-07 19:15:25'
  },
  {
    ID: '7165a61c-a59c-44cd-9d9c-f8b68d14e645',
    Type: 'Placement',
    Message: 'Meta Platforms Inc. hiring',
    Timestamp: '2026-05-07 18:15:21'
  },
  {
    ID: '...',
    Type: '...',
    Message: '...',
    Timestamp: '...'
  },
  {
    ID: '...',
    Type: '...',
    Message: '...',
    Timestamp: '...'
  },
  {
    ID: '...',
    Type: '...',
    Message: '...',
    Timestamp: '...'
  },
  {
    ID: '...',
    Type: '...',
    Message: '...',
    Timestamp: '...'
  },
  {
    ID: '...',
    Type: '...',
    Message: '...',
    Timestamp: '...'
  },
  {
    ID: '...',
    Type: '...',
    Message: '...',
    Timestamp: '...'
  }
]
```

Auth and Token



The screenshot shows a Postman interface with a 'Test Results' tab selected. The response is a JSON object containing user authentication details. The status is 200 OK, and the response size is 677 B. The JSON data is as follows:

```
{
  "email": "dhawalmannu@gmail.com",
  "name": "dhawal shukla",
  "rollNo": "23102002",
  "accessToken": "MdprhE",
  "clientId": "9944f1bd-300c-47dc-9799-0a8216e73010",
  "clientSecret": "vWWHygdXzPVEqzCJ"
}
```

```
{
  "email": "dhawalmannu@gmail.com",
  "Name": "dhawal shukla",
  "rollNo": "23102002",
  "accessToken": "MdprhE",
  "clientId": "9944f1bd-300c-47dc-9799-0a8216e73010",
  "clientSecret": "vWWHygdXzPVEqzCJ"
}
```

2)

